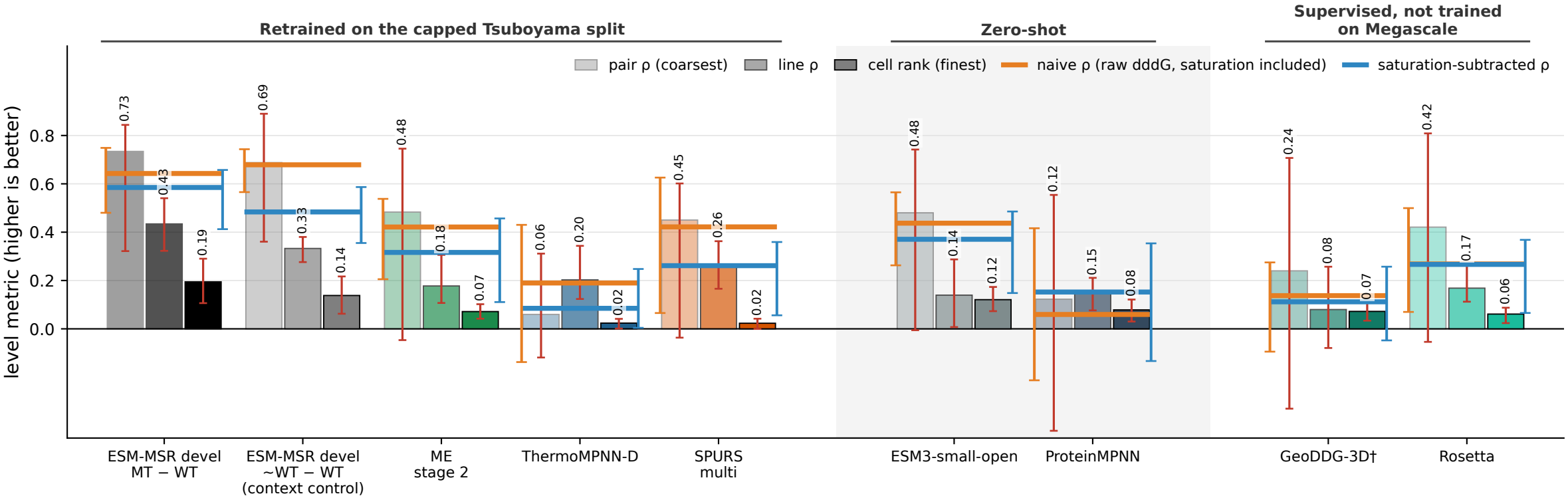
Validation: epistasis predicted level by level, 8,494 doubles with both singles measured in 9 domains



† trained on S8754 (ProThermDB + ThermoMutDB), not on Megascale. ProSST has no interaction term; ESM3ΔG is left out (overlap with its MGnify fine-tune).
Epistasis = the double-mutant prediction of one model minus the additive reading of another or of another head: ESM-MSR MT − WT and ~WT − WT; ME stage 2, ThermoMPNN-D, SPURS multi minus the same seed's singles; ESM3-small-open independent-masking mutant pass minus masked-marginal WT pass; ProteinMPNN, GeoDDG-3D, Rosetta minus the sum of their singles. Seed means over the three seeds of the retrained models.
Intervals: 95% bootstrap over domains (the same resample for every model; the epistasis suite recomputed on each); the whiskers on the dashes (orange left, blue right) are their intervals.